

B. Sc. 5th Semester (Honours) Examination, 2025

Computer Science

Course ID: 51517

Course Code: SH/CSC/504/DSE-2

Microprocessor (Theory) (NEW CBCS Syllabus 2022-23)

Time: 1 Hour 15 Mins

Full Marks: 25

The figures in the margin indicate full marks

Answer all the questions.

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 1 = 5)

- a) What is microprocessor?
- b) What is microcontroller?
- c) Define instruction cycle.
- d) Define machine cycle.
- e) Give the bit position of various flags in flag register of 8085 microprocessor.
- f) Define T- state.
- g) What is Addressing mode?
- h) What do you mean by interrupt?

UNIT II

2. Answer any 2 (two) of the following questions:

(2 × 5 = 10)

-  With timing diagram, explain the memory read operation in 8085 microprocessor.

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- b) Explain the Instruction set of 8085 microprocessor.
- c) Explain the different types of addressing modes of 8085 microprocessor.
- d) What is interfacing? Write the difference between memory mapped I/O and peripheral mapped I/O. 1+4

UNIT III

3. Answer any 1 (one) of the following questions:

1×10=10

- a) Draw and explain the architecture of 8085 microprocessor.
- b) Explain the pin diagram of 8085 microprocessor.

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Answer all the questions.

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 1 = 5)

- a) What is Pixel?**
- b) What is image Enhancement?**
- c) what is image Restoration?**
- d) What are the modes of JPEG?**
- e) What are the three types of discontinuity in digital image?**
- f) What is morphological image processing?**
- g) What is Sampling in image?**
- h) What is geometric transformation?**

[Turn Over]

UNIT II

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2. Answer any 2 (two) of the following questions:

(2 × 5 = 10) C

- a) Write a short note on HIS color image model.
- b) Explain Fourier transformation.
- c) Write short note on dilation and erosion. T
- d) What is redundancy? Explain different types of redundancy in digital image processing 1+4

UNIT III

3. Answer any 1 (one) of the following questions:

1

1×10=10

- a) What are the components of digital image processing system? Explain each in detail.
- b) Explain Boundary descriptors in detail with a neat diagram.

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Microprocessor (Theory)
(OLD CBCS Syllabus 2017-18)

Time: 1 Hour 15 Mins

Full Marks: 25

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Answer all the questions.

UNIT I

1. Answer any 5 (five) of the following questions: (5 × 1 = 5)

- a) What is the difference between the microprocessor and microcontroller?
- b) What is Accumulator?
- c) What is linker?
- d) What are the three classifications of 8086 interrupts?
- e) What are the various programmed data transfer methods?
- f) What is the function of RESET pin of 8086 microprocessor?
- g) What is the function of READY pin of 8086 microprocessor?
- h) What is Assembler?

[Turn Over]

UNIT II

2. Answer any 2 (two) of the following questions: (2 × 5 = 10)

- a) Write down the features of 8086 microprocessor.
- b) Explain the various status flags of 8086 microprocessor.
- c) Write a short note on Write Cycle Timing Diagram for minimum mode.
- d) Write assembly language program for 8086 microprocessor to compute two 8-bit multiplication.

UNIT III

3. Answer any 1 (one) of the following questions 1×10=10

- a) Explain the bus interface unit and execution unit of 8086 microprocessor.
- b) Discuss the various addressing modes of 8086 microprocessor.

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Full Marks: 25

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Answer all the questions.

UNIT I

1. Answer any 5 (five) of the following questions:

(5 × 1 = 5)

- a) What is Digital image?
- b) What are different arithmetic operations applicable to images?
- c) What is Gray level?
- d) Define Resolution.
- e) What is image compression?
- f) Define brightness.
- g) What is an image transform?
- h) What is morphological image processing?

[Turn Over]

UNIT II

2. Answer any 2 (two) of the following questions:

(2 × 5 = 10)

- a) Write a short note on RGB color model.
- b) What is Sampling? Explain it.
- c) Write a short note on image degradation.
- d) Discuss edge detection process in image segmentation.

UNIT III

3. Answer any 1 (one) of the following questions:

1×10=10

- a) Describe the fundamental steps involved in digital image processing system.
- b) Explain wavelet transform and write its application.

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Digital Image Processing (Practical)
(NEW CBCS Syllabus 2022-23)

Time: 2 Hrs

Full Marks: 15

The figures in the margin indicate full marks

(Experiment = 10, LNB + Viva Voce= 05)

Perform any one experiment: (1 × 10 = 10)

- a) Write and execute program for geometric transformation of image
 - (i) Translation
 - (ii) Rotation
- b) Write and execute image processing programs using point processing method
 - (i) Obtain Negative image
 - (ii) Contrast stretching
- c) Write a program (in MATLAB) to implement Sharpening (Highpass) Frequency Domain Filters.
- d) Write a program (in MATLAB) to implement color conversion for HIS model to RGB model.
- e) Write a program (in MATLAB) to perform Histogram Equalization.

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- f) Write and execute program for wavelet transform on an image and perform inverse wavelet transform to reconstruct image.

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Microprocessor (Practical)
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Time: 2 Hrs

Full Marks: 15

The figures in the margin indicate full marks

(Experiment = 10, LNB + Viva Voce = 05)

Perform any one experiment:

(1 × 10 = 10)

- a) Write an assembly language program for 16-bit binary division and multiplication.
- b) Write an assembly language program to add and subtract two arrays.
- c) Write an assembly language program for Bubble sort of an array of numbers.
- d) Write an assembly language program for HCF and LCM of two given integers.
- e) Write an assembly language program for linear search and binary search.
- f) Write an assembly language program for binary to ascii conversion.

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Digital Image Processing (Practical)
(OLD CBCS Syllabus 2017-2018)

Time: 2 Hrs

Full Marks: 15

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(Experiment = 10, LNB + Viva Voce= 05)

1. Perform any one experiment: (1 × 10 = 10)

- a) Write a program (in MATLAB) to implement Arithmetic and logical operation:
 - (i) Image Subtraction
 - (ii) Image Averaging
- b) Write a program (in MATLAB) to implement different Noise model:
 - (iii) Uniform Noise
 - (iv) Salt & Pepper Noise
- c) Write a program (in MATLAB) to implement color conversion for GB model to HIS model.
- d) Write a program (in MATLAB) to perform smoothing and sharpening on color image.
- e) Write and execute program for image morphological operations erosion and dilation.

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**Microprocessor (Practical)
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Time: 2 Hrs

Full Marks: 15

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(Experiment = 10, LNB + Viva Voce= 05)

1. Perform any one experiment: (1 × 10 = 10)

- a) Write an assembly language program to find the largest number from an array of data.
- b) Write an assembly language program for ascii to binary conversion.
- c) Write an assembly language program for Linear search and binary search.
- d) Write an assembly language program to add and subtract two arrays.
- e) Write an assembly language program for 32-bit binary division and multiplication.
- f) Write an assembly language program for binary to ascii conversion.

[Turn Over]

- f) Write a program (in MATLAB) to perform Histogram Equalization.